

Executive Summary – Customer Churn Analysis

This report presents a comprehensive analysis of customer churn within a telecommunications company. The goal is to identify key drivers of churn and provide actionable insights for retention strategies. The dataset contains 7,043 entries and includes customer demographic information, service subscriptions, billing methods, and churn status.

◆ Overall Churn Overview

- The **overall churn rate** is **26.5%**, indicating that roughly **1 in 4 customers** left the service.
 - **73.5%** of customers remained with the company, showing potential loyalty that can be further leveraged.
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◆ Demographic Insights

- **Senior Citizens:**
 - Comprise **16.2%** of the customer base.
 - Among Senior Citizens, the **churn rate is 41.9%**, significantly higher than the **24.0% churn rate** of non-senior customers.
 - This suggests age-specific strategies (e.g., more support for older users) could reduce churn.
 - **Gender:**
 - Males and females are almost equally represented and show similar churn behavior:
 - **Female churn rate:** 26.5%
 - **Male churn rate:** 26.1%
 - Gender does not appear to be a major differentiator in churn behavior.
 - **Partner and Dependents:**
 - **Customers without partners** have a churn rate of **32.2%** vs. **20.3%** for those with partners.
 - **Customers without dependents** churn at **31.0%**, compared to only **16.1%** for those with dependents.
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◆ Tenure & Contract Duration

- Tenure is a major driver of churn:
 - **Tenure $\leq$ 1 year:** churn rate is **45.5%**
 - **Tenure $\geq$ 5 years:** churn rate drops drastically to **6.3%**

- Customers with longer tenure are more loyal.
- Contract type heavily influences churn:
 - **Month-to-month: 43.9% churn rate**
 - **One-year contract: 11.0% churn rate**
 - **Two-year contract: only 2.8% churn rate**
 - Suggests that long-term contracts significantly reduce churn risk.

◆ Service Subscription Analysis

Churn is influenced by the types of services customers subscribe to:

Service	Churn Rate (Yes)	Churn Rate (No)
PhoneService	26.2%	25.0%
MultipleLines	30.4%	24.9%
InternetService	30.6% (DSL/Fiber)	7.5% (No internet)
OnlineSecurity	15.2%	41.1%
OnlineBackup	21.6%	32.5%
DeviceProtection	22.5%	32.2%
TechSupport	14.0%	39.5%
StreamingTV	26.6%	26.3%
StreamingMovies	27.0%	25.7%

- Lack of **security and support services** like OnlineSecurity and TechSupport correlates with **higher churn**, emphasizing the need to promote and bundle these services.

◆ Payment and Billing Insights

- **Payment Method:**
 - **Electronic check** users have the highest churn rate at **45.5%**.
 - Other methods:
 - **Mailed check:** 19.6%
 - **Bank transfer (auto):** 16.7%
 - **Credit card (auto):** 15.3%
- **Paperless Billing:**

- Customers with paperless billing churn at **33.8%**, vs. only **16.1%** for those with mailed bills.
 - **Monthly Charges:**
 - Customers paying **above \$80/month** are more likely to churn (around **40%**), whereas those paying **less than \$50/month** have a churn rate of only **15%**.
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◆ Data Preprocessing and Feature Engineering

- **Missing values:** Found and replaced in the TotalCharges column using zero-fill for blank spaces, then converted to numerical.
 - Binary columns such as SeniorCitizen were mapped to categorical values ('Yes'/'No') for better interpretability.
 - Categorical features were evaluated for group-wise churn distribution using bar plots.
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◆ Visualizations and Dashboard

- A variety of **count plots**, **stacked bar charts**, and **pie charts** were used.
 - Each categorical service column was visualized in a **subplot grid layout** to allow for quick comparative analysis.
 - Percentage labels were added for better understanding of proportions, especially in churn-vs-non-churn charts.
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◆ Key Takeaways

- Target retention strategies toward:
 - **Month-to-month contract holders**
 - **Customers paying via electronic checks**
 - **Low-tenure customers (≤ 1 year)**
 - **Customers lacking support/security services**
- Encourage:
 - **Longer contract commitments**
 - **Bundling tech support or security**
 - **Switching to auto-payment methods**
- Focus churn-reduction campaigns on:
 - **Senior citizens**
 - **Customers without dependents**

- **High-bill customers (>\$80)**